CashJournal — Complete Application Audit & Feature Reference (v5.0)

Exhaustive architectural, mathematical, cryptographic, UI/UX, CSV import, recurring bills, OTA update, and security audit through Phase 30 (v1.0.36-dev • Updated 2026-10-17 19:12 UTC+07:00).

Audit Revision v5.0 (2026-10-17 19:12 UTC+07:00): Updates Audits v1.0 (v1.0.14-dev), v2.0 (v1.0.18-dev), v3.0 (v1.0.28-dev), and v4.0 (v1.0.35-dev) to cover Phase 30 (v1.0.36-dev): elimination of destructive migration fallback (OBS-NEW-4), multi-currency conversion for recurring bills (OBS-NEW-2), full reset wiping recurring bills (OBS-NEW-1), CSV category.key normalization (OBS-NEW-3), Ground-Truth Code Verification Matrix answering all 8 external review questions, resolution of all 26 prior findings & observations (26/26), and 17 JUnit4 test suites (120 unit tests — 100% PASS).

1. Chronological Audit & Engineering Time Log (UTC+07:00)

Timestamp (+07:00)	Version / Commit	Phase / Milestone	Summary of Changes & Audit Events
2026-10-15 22:00	d08e35a (v1.0.9)	Phase 8 Stabilization	Fixed onboarding permission launcher crash (fragment-ktx:1.8.1), clarified storage sandbox, refined 4-dot signal cluster.
2026-10-15 23:17	54d66d5 (v1.0.10)	Brand Identity	Integrated official CashJournal Logo v1 across adaptive launcher icons and in-app branding.
2026-10-16 12:52	91f3803 (v1.0.11-dev)	Sensory & i18n	Bilingual localization groundwork, knock/shake detector calibration, and audio chime synthesis.
2026-10-16 13:20	337ba09 (v1.0.12-dev)	Lock & Layout	Calibrated text hierarchy, layout geometry, 15s background auto-lock lifecycle, and wave mask performance.
2026-10-16 14:09	f1ce18c	Sub-Phase 9A	Room v1->v2 migration (initialBalance), zero-drift SQL derivation, backdating, full CRUD, and balance reconciliation.
2026-10-16 14:22	10b8967	Sub-Phase 9B	Versioned JSON backup export & atomic restore via Storage Access Framework (BackupManager).
2026-10-16 14:49	10c5ec1 (v1.0.13-dev)	Sub-Phase 9C	Custom payday cycle start day (1..28), expanded 12 preset categories, and manual category picker modal.
2026-10-16 15:43	28eaece (v1.0.14-dev)	Pre-Audit Baseline	Synchronized BuildConfig.VERSION_NAME display in Settings.
2026-10-16 15:54	v1.0.14-dev	Audit Report v1.0	Initial full-codebase audit completed. Identified 16 issues (1 Critical, 4 High, 5 Medium, 6 Low).
2026-10-16 16:21	7bb6203	Phase 10 Remediation	Fixed CRITICAL-1 (initialBalance currency conversion), HIGH-1 (live selectedAccount), HIGH-2 (world currency sync), HIGH-4 (cycle startingSpent), LOW-5 (New Year & DST math).
2026-10-16 16:31	5dcca0d	Phase 11 Remediation	Fixed HIGH-3 (weather Off hides widget), MED-1 (word-boundary regex \b in Category), MED-2 (buffered budget slider), MED-3 (SALARY picker & category/accent preservation).
2026-10-16 17:08	7259a0b (v1.0.15-dev)	Phase 12 Remediation	Fixed MED-4 (10,000-round salted SHA-256 PIN + PinSecurityTest), MED-5 (full ID localization), LOW-1 (18-pref backup), LOW-2 (DailyReminderScheduler 20:00), LOW-3/4/6.
2026-10-16 17:49	3edae55 (v1.0.16-dev)	Phase 13 Polish	Added isPreferencesLoaded cold-start gate (eliminating Frame-0 onboarding flash) and centered spring morph between Status Ring and Back Button.
2026-10-16 18:37	19ff083 (v1.0.17-dev)	Phase 14 Features	Added sticky immersive fullscreen, interactive PhoneSilhouettePreview in Onboarding Step 7, auto-triggered ReleaseNotesSheet, and reactive tooltip localization.
2026-10-16 19:31	2c112d3 (v1.0.18-dev)	Phase 15 Polish	Fixed AccountDetail back button closure, 2-row balance card + auto-scaling MaskedAmount, natural EN/ID copywriting overhaul, 1:2 phone silhouette (118x236dp), inline knock sensitivity.
2026-10-16 19:42	v1.0.18-dev	Audit Report v2.0	Post-Phase-15 re-audit completed. Verified 16/16 initial findings resolved and 10/10 JUnit test suites passing.
2026-10-16 20:44	95927db (v1.0.19-dev)	Phase 16A/B OTA & Types	Added AccountType (Room v2->v3 MIGRATION_2_3), EditAccountSheet, and GitHub OTA In-App Update System (AppUpdateService, UpdateSheet, FileProvider).
2026-10-16 22:22	7c46501 (v1.0.20-dev)	Phase 16C Lock Veil	Moved isAppLocked/isVeiled into MainViewModel; engaged immediate privacy veil on pause while suppressing biometric prompt on minimize.
2026-10-16 23:13	3566a10 (v1.0.21-dev)	Phase 17 Hardening	Added AppVersionProvider (PackageManager query), locked ModalBottomSheet guard, and 60s isInstallingUpdate grace window with OEM installer URI grants.

Timestamp (+07:00)	Version / Commit	Phase / Milestone	Summary of Changes & Audit Events
2026-10-16 23:37	321e31f (v1.0.22-dev)	Phase 18 Hygiene	Added cleanStaleUpdateApks() and purgeOtherUpdateApks() in AppUpdateService to automatically reclaim cache storage.
2026-10-16 23:44	a8e7b54	Phase 19 Docs	Added Play-Store-style README.md and MIT LICENSE.
2026-10-17 00:28	1acd4e0 (v1.0.23-dev)	Phase 20 Brands	Added BrandRegistry (42 Indonesian & global banks/e-wallets), 16 curated color swatches, CustomHexColorDialog (#RRGGBB), and radial color haze on account cards.
2026-10-17 00:57	eb01527 (v1.0.24-dev)	Phase 21 Runway	Upgraded Room v3->v4 (MIGRATION_3_4) with BudgetMode (FIXED_NOMINAL vs PERCENTAGE_OF_FUNDS), budgetPercent, and Capital Burn-Rate Survival Runway.
2026-10-17 10:51	0ea309b (v1.0.25-dev)	Phase 22 UX & FX	Expanded % of Total Funds slider to 1%..100%, widened Dashboard carousel cards (138dp), added ThousandsSeparatorVisualTransformation, and anchored FX to /v6/latest/USD.
2026-10-17 11:36	859d244 (v1.0.26-dev)	Phase 23 Safeguards	Added disk-aware ReadyToInstall promotion, 1-tap update sheet from glowing Settings icon, and non-destructive onboarding replay safeguards.
2026-10-17 12:54	74440da (v1.0.27-dev)	Phase 24 Island & Dock	Engineered detached Liquid Floating Status Ring Island on nested screens, Thumb-Zone Floating Action Dock on AccountsScreen, unified EditAccountSheet, App Updates card, and USAGE_MEDIA chimes.
2026-10-17 13:14	72793a8 (v1.0.28-dev)	Phase 25 Halo & Notes	Unified Liquid Status Island width (54dp), added Cobalt Breathing & Ripple Update Halo on Settings tab, added Android 11+ installer <queries>, and curated bilingual ReleaseNotesSheet.
2026-10-17 13:25	v1.0.28-dev	Audit Report v3.0	Comprehensive re-audit through Phase 25 completed. Verified 14 JUnit4 test suites (91 unit tests) passing and logged OBS-1..OBS-6.
2026-10-17 13:45	bb62d72	Phase 26-29 Roadmap	Logged external review notes (docs/cashjournal_review_notes_from_Claude.md) and architected Phases 26-29 in PROJECT_LOG.md.
2026-10-17 14:14	8b4c8cd (v1.0.29-dev)	Phase 26 Hygiene & 30d Runway	Resolved OBS-1..4, OBS-6: split BootReceiver (exported=true) from DailyReminderReceiver (exported=false); removed dead sheets/previews; added 30-day rolling burn window + 5-day calibration threshold.
2026-10-17 14:44	e18eeb7 (v1.0.30-dev)	Phase 27 Recurring Bills	Upgraded Room v4->v5 (MIGRATION_4_5, recurring_transactions); added RecurringBillsSheet, Dashboard Upcoming Bills card, Safe-to-Spend committedBills deduction, and 1-tap Mark as Paid.
2026-10-17 15:32	f6c8b26 (v1.0.31-dev)	Hotfix Startup Guards	Added fallbackToDestructiveMigration(), MainViewModel .catch flow guards, and API 28+ / Throwable guards on TelephonyManager.signalStrength.
2026-10-17 15:49	3e07abc (v1.0.32-dev)	Hotfix DEX 255-Register Fix	Resolved ART java.lang.VerifyError by refactoring ScreenStrings from a 301-param data class into an interface with zero-arg property getters; added MigrationTest.kt (7 SQLite JDBC tests).
2026-10-17 16:14	a401da8 (v1.0.33-dev)	Hotfix Ledger Isolation	Enforced destinationAccountId = null on non-TRANSFER transactions across ViewModel/Sheets plus startup SQL sanitizeNonTransferDestinations().
2026-10-17 16:50	e87a6bf (v1.0.34-dev)	Phase 28 CSV Import/Export	Added CsvTokenizer, CsvParser (BCA CR/DB, Mandiri Debit/Kredit, BNI/BRI, US/EU), CsvDuplicateDetector, CsvExporter, CsvImportSheet, AccountDetail 'Import CSV', and CsvParserTest.kt (6 tests).
2026-10-17 17:35	3ade377 (v1.0.35-dev)	Phase 29 AES-256 & R8	Resolved OBS-5: added BackupCrypto (AES-256-GCM + PBKDF2 100k + CJENC1 header), encrypted .enc export/restore, R8 minification + proguard-rules.pro, release keystore support, and BackupCryptoTest (5 tests).
2026-10-17 17:41	v1.0.35-dev	Audit Report v4.0	Complete codebase audit through Phase 29 completed. Verified all 22 prior findings/observations resolved, 17 JUnit4 test suites (118 tests) passing, and logged OBS-NEW-1..4.
2026-10-17 18:15	61b5f96 (v1.0.36-dev)	Phase 30 Data Integrity	Resolved OBS-NEW-1..4: removed fallbackToDestructiveMigration(); added multi-currency conversion for recurring_transactions.amount; added recurringTransactionDao().deleteAll() to clearAllData(); normalized CSV category.key; added 2 tests in LedgerIntegrityTest (120 total).
2026-10-17 19:12	v1.0.36-dev	Audit Report v5.0	Complete codebase audit through Phase 30 completed. Verified all 26 prior findings/observations resolved, answered all 8 external review questions, and verified 17 JUnit4 test suites (120 unit tests) passing.

2. Executive Summary & Architectural Overview

CashJournal (app.cashjournal, current build v1.0.36-dev, monotonic autoVersionCode) is a native, offline-first personal finance, multi-account ledger, recurring bills manager, bank statement importer, and capital runway application for Android

(minSdk 26 through targetSdk 34). Built with Kotlin 2.0, Jetpack Compose, Room SQLite v5, Preferences DataStore, Kotlinx Serialization, OkHttp 4.12, and Phosphor Icons, the app is engineered around five non-negotiable pillars:

- 100% Free & Keyless: Zero paid libraries, zero user accounts, and zero API keys (Open-Meteo, ExchangeRate-API Open Access /v6/latest/USD, and GitHub Releases API).
- 100% Local, Transparent & Cryptographically Protected: All financial records reside in the internal app sandbox (allowBackup="false"). User-initiated backups support authenticated AES-256-GCM encryption (PBKDF2WithHmacSHA256, 100,000 iterations, 16B random salt, 12B IV, 128-bit auth tag, CJENC1 binary header) as well as plaintext JSON and CSV spreadsheet exports.
- Zero-Drift & Zero-Loss Mathematical Integrity: Currency values stored as 64-bit integer minor units (Long); balances derived via SQL initialBalance + SUM(inflows) - SUM(outflows) across manual entries, retroactive edits, balance reconciliations, recurring bill payments, currency conversions, and CSV imports. Forward database upgrades are strictly non-destructive (no fallbackToDestructiveMigration).
- Dual-Layer Network Lockdown & Manifest Isolation: Both network_security_config.xml and HostAllowlistInterceptor restrict HTTPS traffic to 6 hosts. Broadcast receivers are split between BootReceiver (exported=true for BOOT_COMPLETED) and DailyReminderReceiver (exported=false).
- Ergonomic One-Handed UI & Sensory Feedback: Detached 54.dp Liquid Floating Status Ring Island, Thumb-Zone Floating Action Dock, Cobalt Breathing & Ripple Update Halo, accelerometer knock/shake side-swap (ShakeDetector), 44.1kHz PCM chimes on USAGE_MEDIA (ChimeGenerator), and DEX-safe 325-key bilingual interface localization (AppStrings).

3. Complete Feature & Functional Reference (v1.0.36-dev)

3.1 Entry Point, Privacy Veil, DEX-Safe i18n & Self-Healing Startup (MainActivity.kt & MainViewModel.kt)

- Cold-Start Initialization Gate: Holds a clean Slate100 surface on Frame 0 until state.isPreferencesLoaded is true, eliminating any flash of OnboardingScreen before DataStore finishes reading.
- Zero-Latency Background Privacy Veil: Immediately triggers onAppBackgrounded() in onPause() to veil all sensitive content behind AppLockScreen, while deferring the biometric prompt until the user returns after >=15s (autoPromptBiometrics = isAppLocked && !isVeiled). Includes a 60s isInstallingUpdate grace window during OTA APK upgrades.
- Startup Self-Healing Ledger Isolation: Runs TransactionDao.sanitizeNonTransferDestinations() on startup to clear any legacy non-null destinationAccountId on EXPENSE/INCOME rows.
- DEX-Safe Localization Interface (AppStrings.kt): Implements ScreenStrings as a Kotlin interface (325 properties) with singleton EnglishScreenStrings and IndonesianScreenStrings classes, avoiding Android ART's 255-register constructor VerifyError.

3.2 Primary Screens (7 Screens) & Modal Bottom Sheets (12 Sheets)

Screen / Sheet	Key Features, Telemetry & Interactive Flows (v1.0.36-dev)
DashboardScreen	Total Net Worth hero; 28.dp Safe-to-Spend Today card with 64.dp % gauge, status pill, 30-Day Rolling Survival Runway pill (or Saffron 'Calibrating... (X days needed)' when history <5d), and Committed Bills deduction subline; 138.dp Active Accounts carousel; Upcoming Bills card with 'Manage' action; Recent Activity; Monthly Budget Runway card.
LedgerScreen	Transactions ('Riwayat Transaksi') with 4 live-count filter pills (All, Expense, Income, Transfer); instant search (notes, EN/ID categories, raw digits); chronological day grouping (TODAY, YESTERDAY, date headers).
AccountsScreen	World Currencies card synced with USD-anchored cross-rates; 24.dp account cards with radial brand color haze, AccountType icon + pill badge, % share pill, and spring animateItemPlacement; bottom Thumb-Zone Floating Action Dock ('Reorder' + '+ Add').
AccountDetailScreen	Live-updating single-account view; detached 54.dp Liquid Status Island + Back Button in SideRail; Balance Card with 'Manage' button (opening EditAccountSheet) AND 'Import CSV' button (opening SAF CSV picker pre-targeted to this account); Money In & Money Out tiles; strictly isolated account activity feed.
AnalyticsScreen	Last 6 Months Money In / Money Out bar chart; Dual-Mode Monthly Budget Calibrator ('Fixed Amount' vs '% of Total Funds' 1%..100% slider) + 30-Day Rolling Capital Survival Runway card (with 5-day calibration threshold notice); cycle-filtered Top Spending Categories Donut Chart.
SettingsScreen	12 cards: Language (EN/ID), Main Currency, Monthly Reset Day (1..28), Recurring Bills (Manage Recurring Bills), Security & Access (PIN, Biometric, FLAG_SECURE, Lock Now), Sensory & Feedback (Sound, Haptics, 20:00 Reminder), Menu Position, Weather, Data & Backup (Export Encrypted .enc, Export Plaintext JSON, Restore Data with auto-decrypt prompt, Import Bank CSV, Export Spreadsheet CSV), App Updates, About, and Developer/Danger Zone.
AppLockScreen	Full-screen lock screen & background privacy veil verifying 4-digit PIN via 10,000-round salted SHA-256 or BiometricPrompt.
RecurringBillsSheet	Phase 27 & 30: List view with Total Committed summary card, status badges (Paused, Paid this cycle, Due today, Due in Xd, Day X), active toggle, and 1-tap 'Mark as Paid'; Add/Edit form supporting Monthly (1..28) and Weekly (Mon..Sun) bills. Automatically converted on main currency switch and wiped on full reset.

Screen / Sheet	Key Features, Telemetry & Interactive Flows (v1.0.36-dev)
CsvImportSheet	Phase 28 & 30: Preview sheet showing Total Rows, Selected (Emerald), and Duplicates (Saffron), live target account switcher (re-running duplicate detection), Select/Deselect All, and per-row checkboxes with normalized Category keys (item.category.key).
10 Other Sheets	LogTransactionSheet, TransactionEditSheet, NewAccountSheet (42 brands + 16 swatches + #RRGGBB), EditAccountSheet (unified edit + live thousands-formatted balance reconciliation + delete guard), MainCurrencySheet (localized + converts recurring bills), WorldCurrenciesSheet, WeatherDetailSheet, PinSetupSheet, UpdateSheet, and ReleaseNotesSheet (v1.0.36-dev highlights).

4. Data Architecture, Cryptography, Formulas & Automated Test Suite

4.1 Room v5 Schema, Committed Bills, 30-Day Rolling Runway & AES-256-GCM Backups

```
balance = initialBalance + SUM(INCOME src) + SUM(TRANSFER dst) - SUM(EXPENSE src) - SUM(TRANSFER src)
remaining = max(0, effectiveBudget - totalSpent - committedBills) | safePerDay = floor(remaining / max(1, daysLeft))
dailyBurnRate = floor(rolling30dExpenseSum / min(30, effectiveHistoryDays)) [gated by effectiveHistoryDays >= 5]
ENC Header = 'CJENC1' (6B) || iterations_be32 (100,000) || salt (16B) || iv (12B) || AES-256-GCM(UTF8_JSON) + 16B Tag
```

- Strict Non-Destructive Room v1 -> v5 Migrations (AppDatabase.kt & MigrationTest.kt): Deterministic migrations add initialBalance (v1->v2), accountType with SQL keyword inference (v2->v3), budgetMode + budgetPercent (v3->v4), and recurring_transactions with 15 columns and 3 indices (v4->v5). fallbackToDestructiveMigration() is completely removed.
- Multi-Bank CSV Statement Parser (CsvParser.kt & CsvDuplicateDetector.kt): Auto-detects comma/semicolon delimiters, BCA single-column CR/DB suffixes, Mandiri/BNI/BRI Debit/Kredit columns, and US/EU decimals across 13 date formats + Indonesian month names (Mei, Agt, Okt, Des). Flags duplicates matching amount + type + same day/24h + title similarity.
- Release Build & R8 Hardening (app/build.gradle.kts & proguard-rules.pro): Enables isMinifyEnabled = true on release builds (reducing release APK size by 36% from 12.4 MB to 7.8 MB), monotonic autoVersionCode, and external keystore support via Gradle properties or environment variables.

4.2 Automated Unit Test Suite Status (17 Test Suites, 120 Tests — 100% PASS)

Test Suite	Tests	Verified Assertions & Coverage	Result
AcceptanceCheckTest	7	Build Brief Section 16 acceptance numbers (Rp 18.45M net worth, Rp 150.294/day on Oct 14, 66% ring, Rp 1M expense -> Rp 91.470/day & 79% Near Limit, transfer isolation, knock sensitivity, 7-dot mask).	PASS
AccountTypeTest	4	Verifies AccountType.fromKey safe fallback, inferFromName classification across Indonesian & English bank/e-wallet/savings/card names, and bilingual titles.	PASS
AppUpdateTest	4	Verifies semantic version comparison (v-prefix, -dev suffix) and APK cache hygiene (cleanStaleUpdateApks, purgeOtherUpdateApks).	PASS
BackupCryptoTest	5	Verifies AES-256-GCM encrypt/decrypt round-trip, wrong-password DecryptionFailedException, tampered ciphertext GCM tag failure, isEncrypted header detection, and truncated header rejection.	PASS
BackupRestoreTest	7	BackupPayload JSON round-trip (accounts, transactions, monthlyPlan, 22 prefs, recurringTransactions), encrypted stream round-trip, inspectBackup detection, and zero-drift recalculation.	PASS
BrandRegistryTest	10	Verifies auto-detection across 42 Indonesian/global banks & e-wallets (BCA, Mandiri, Jago, SeaBank, GoPay, OVO, Wise, Amex), word-boundary anti-collision, and custom #RRGGBB hex parsing.	PASS
CategoryDetectorTest	9	Keyword detection across categories plus word-boundary negative tests confirming 'theater', 'chair', and 'cost' do not trigger false positives.	PASS
CsvParserTest	6	Verifies BCA CR/DB format with SALDO AWAL filtering, Mandiri Debit/Kredit with Indonesian months, quoted multiline CSV fields, EU semicolon/comma decimals, duplicate detection, and CsvExporter.	PASS
CurrencyConverterTest	7	Same-currency identity, USD <-> IDR decimal scaling (2 vs 0 decimals), EUR -> JPY cross-rates, USD-anchored high-precision IDR conversion, and missing-rate fallback.	PASS
HostAllowlistTest	3	Verifies HostAllowlistInterceptor permits the 6 allowed hosts (Open-Meteo, ExchangeRate-API, GitHub Releases/CDN) and throws SecurityException on unauthorized domains.	PASS
LedgerIntegrityTest	7	Expanded in Phase 30 (+2 tests): Zero-drift derivation on retroactive edits/deletes, atomic transfer conservation, reconciliation delta math, backdating, initialBalance currency conversion, multi-currency recurring bill conversion, and 4-table full reset wipe.	PASS
MigrationTest	7	Real SQLite JDBC migrations v1->v2, v2->v3, v3->v4, v4->v5, and full v1->v5 chain, verifying data preservation and exact schema parity with Room v5.	PASS
PinSecurityTest	5	Verifies saltHex:hashHex format (32:64 hex chars), per-call salt randomness, 10,000-round salted verification, legacy 64-hex unsalted fallback, and malformed hash safety.	PASS

Test Suite	Tests	Verified Assertions & Coverage	Result
SafeToSpendTest	23	Leap-year Feb, custom cycleStartDay, PERCENTAGE_OF_FUNDS budget, 5-day calibration threshold, trailing 30-day rolling burn rate across payday resets, and committedBills deduction.	PASS
SignalStrengthHelperTest	3	Monotonic translation of 4-bar and 5-bar cellular signal scales into 4 UI dots.	PASS
ThousandsSeparatorTest	5	Verifies Indonesian dot grouping, US comma grouping, empty/short strings, and bidirectional cursor OffsetMapping.	PASS
WeatherIconMapperTest	8	WMO weather code mapping across day/night variants to Phosphor icons.	PASS

5. Audit Findings Resolution Ledger & Ground-Truth Verification Matrix

5.1 Resolution of All Prior Audit Findings & Observations (26 / 26 Resolved)

ID	Origin	Prior Finding / Observation Summary	Resolution Status & Verification
CRITICAL-1	Audit v1.0	changeMainCurrency() omitted acc.initialBalance conversion.	RESOLVED (Phase 10) — Converts initialBalance + recalculates all balances; verified in LedgerIntegrityTest.
HIGH-1..4	Audit v1.0	Stale selectedAccount, world currency sync, weather Off state, custom cycle startingSpent.	RESOLVED (Phases 10-11, 22) — Live state derivation, USD-anchored FX, hidden weather Off state, cycleMonthKey.
MED-1..5	Audit v1.0	Category regex \b, buffered slider, SALARY picker, 10k salted SHA-256 PIN, ID localization.	RESOLVED (Phases 11-12) — Verified across CategoryDetectorTest, PinSecurityTest, and 325-key AppStrings.
LOW-1..6	Audit v1.0	Backup prefs, Daily Reminder scheduler, haptics/chime gating, DST/New Year dates, dead code.	RESOLVED (Phases 10-12) — 22-pref backup, 20:00 AlarmManager scheduler, haptics checks, DST rounding.
OBS-1..6	Audit v3.0	EditAccountSheet thousands separator, dead sheet/preview files, BootReceiver split, localized toast/dialog, R8 & release keystore.	RESOLVED (Phases 26 & 29, 8b4c8cd / 3ade377) — All 6 observations fixed and verified.
OBS-NEW-1	Audit v4.0	clearAllData() omitted database.recurringTransactionDao().deleteAll().	RESOLVED (Phase 30, 61b5f96) — Added deleteAll() in MainViewModel.kt:1336; verified in LedgerIntegrityTest.
OBS-NEW-2	Audit v4.0	changeMainCurrency() did not convert recurring_transactions.amount.	RESOLVED (Phase 30, 61b5f96) — Converts all recurring bills in MainViewModel.kt:1299-1318; verified in LedgerIntegrityTest.
OBS-NEW-3	Audit v4.0	confirmCsvImport() saved category.name ('FOOD') instead of category.key ('Food').	RESOLVED (Phase 30, 61b5f96) — Normalized to item.category.key in MainViewModel.kt:1551.
OBS-NEW-4	Audit v4.0	.fallbackToDestructiveMigration() in AppDatabase.kt risked wiping data on schema mismatch.	RESOLVED (Phase 30, 61b5f96) — Removed fallbackToDestructiveMigration() from AppDatabase.kt:144.

5.2 Ground-Truth Code Verification Matrix (Answering External Reviewer Questions)

#	Reviewer Verification Question	Ground-Truth Code Answer & Citation (v1.0.36-dev)
Q1	Is .fallbackToDestructiveMigration() still present in the release build path?	No. Removed in Phase 30 (61b5f96). AppDatabase.kt:138-145 configures only MIGRATION_1_2..4_5 and fallbackToDestructiveMigrationOnDowngrade().
Q2	Does changeMainCurrency() update recurring_transactions.amount?	Yes. MainViewModel.kt:1299-1319 converts all recurring bill amounts via CurrencyConverter.convertMinorUnits() inside database.withTransaction.
Q3	Does clearAllData() clear the recurring bills table?	Yes. MainViewModel.kt:1330-1337 calls database.recurringTransactionDao().deleteAll() inside database.withTransaction.
Q4	How is the PIN stored/verified, and is there rate limiting or lockout?	UserPreferencesRepository.kt:353-398 uses 16B SecureRandom salt + 10,000-round SHA-256 (MessageDigest.isEqual). AppLockScreen.kt:96-116 enforces a 400ms reset delay on wrong PIN; no multi-attempt lockout timer yet.
Q5	Is the Room database encrypted at rest, or only backups?	Exported .cashjournal.enc backups use AES-256-GCM (BackupCrypto.kt). Internal cashjournal_db relies on Android OS File-Based Encryption (FBE) & UID sandbox (allowBackup=false), without SQLCipher.
Q6	How are credit card balances treated in the runway calculation?	MainViewModel.kt:247 sums all accounts (rawAccounts.sumOf { it.balance }) regardless of AccountType and passes totalWorth as totalNetWorth into SafeToSpendCalculator.
Q7	Does onboarding privacy copy say what the app sends over the network?	Yes. OnboardingStrings.kt:84, 102 states: 'Zero financial data leaves your phone. Only approximate coordinates are sent to fetch weather.'
Q8	Do chimes default to on or off?	On (true). UserPreferencesRepository.kt:37, 118 sets soundEffectsEnabled = true by default (toggleable in Settings -> Sensory & Feedback).